Hybrid Three-Dimensional LTD/CFF Representations

Implementation and theory note

April 2026

Abstract

This note describes the derivative-free three-dimensional representation implemented in `hybrid3d.py`. The construction combines loop-tree duality (LTD) with the causal/CFF representation in order to handle repeated propagators while keeping the numerator as a sampled black box. It also describes the bounded-energy-degree extension used when edge-momentum representation (EMR) numerators contain powers above the affine CFF class.

The document is organized as follows. Chapter 1 gives the mathematical construction, with all dependencies made explicit. Chapter 2 explains how the Python implementation and the oriented JSON schema encode the formulae. Chapter 3 reports current Symbolica evaluator timings from the example scripts.

Contents

1 Theoretical Construction **2**
- 1.1 Framing, Notation, LTD, and CFF . . . 2
- 1.2 Repeated Propagators Without Numerator Derivatives . . . 3
  - 1.2.a The Fully Generic Formula . . . 3
  - 1.2.b The Box with a Triple Repeated Propagator . . . 7
- 1.3 Higher Energy Powers in Numerators . . . 9
  - 1.3.1.a Quadratic Bounds . . . 9
  - 1.3.1.b One-Loop Box with One Quadratic Edge . . . 10
  - 1.3.2.a Bounds Above Quadratic . . . 11
  - 1.3.2.b One-Loop Box with One Cubic Edge . . . 13
  - 1.3.2.c What the First Rejected Contact Sector Contains . . . 14

2 Python Implementation and JSON Format **17**
- 2.1 Module Structure . . . 17
- 2.2 Bounded-Degree Support Matrix . . . 17
- 2.3 Oriented JSON Semantics . . . 18
- 2.4 Surface Cache . . . 18
- 2.5 Orientation Labels and Variants . . . 18
- 2.6 Command-Line Workflow . . . 19
- 2.7 Symbolica Evaluators . . . 19

3 Performance **20**
- 3.1 Benchmark Setup . . . 20
- 3.2 Repeated Box . . . 20
- 3.3 One-Loop 10-Gon . . . 20
- 3.4 Five-Loop Four-External Benchmarks . . . 21
- 3.5 Precision Stability . . . 22

Chapter 1

Theoretical Construction

1.1 Framing, Notation, LTD, and CFF

Let $\mathcal{G}$ be a connected scalar graph with L loop momenta and internal edge set E. A loop-momentum basis (LMB) is denoted by $\mathcal{B} = (k_1, \dots, k_L)$. For an edge $e \in E$,

$$q_e(k,p) = \sum_{r=1}^{L} \rho_{er} k_r + \sum_{a=1}^{n_{\text{ext}}} \eta_{ea} p_a, \qquad \rho_{er}, \eta_{ea} \in \mathbb{Z}, \tag{1.1}$$

and

$$q_e^0(k^0,p^0) = \rho_e \cdot k^0 + \eta_e \cdot p^0, \qquad \omega_e(\vec{k},p,m) = \sqrt{\vec{q}_e^{\,2} + m_e^2}. \tag{1.2}$$

The Feynman denominator is

$$D_e = (q_e^0)^2 - \omega_e^2 + \mathrm{i}0. \tag{1.3}$$

For a numerator N in EMR variables, the D-dimensional input integral is

$$I_{\mathcal{G},\mathcal{B}}[N] = \int_{\mathbb{R}^{4L}} \prod_{r=1}^{L} \frac{\mathrm{d}^4 k_r}{(2\pi)^4} N(\{q_e(k,p)\}_{e\in E}) \prod_{e\in E} D_e^{-a_e}. \tag{1.4}$$

The powers a_e are normally one; repeated propagators are encoded below as explicit repeated edges or, equivalently, as collapsed channels with powers.

LTD dependency data. The LTD construction is an algorithmic residue expansion of (1.4) after choosing

- the graph $\mathcal{G}$,
- the loop-momentum basis $\mathcal{B}$,
- a contour-closure prescription κ, i.e. the side on which each loop-energy integration is closed.

We denote the resulting spatial integrand by

$$I^{(3)}_{\text{LTD}}[\mathcal{G},\mathcal{B},\kappa;N](\vec{k};p,m) = \sum_{\mathfrak{o}\in\mathcal{O}_{\text{LTD}}(\mathcal{G},\mathcal{B},\kappa)} N\Big(Q_{\mathfrak{o}}^{\text{LTD}}(\vec{k};p,m)\Big)\, W_{\mathfrak{o}}^{\text{LTD}}(\vec{k};p,m). \tag{1.5}$$

The map $Q_{\mathfrak{o}}^{\text{LTD}}$ gives every EMR edge energy after solving the cut equations. The weight $W_{\mathfrak{o}}^{\text{LTD}}$ is a product of cut half-edge factors and LTD dual surfaces. In this note we do not rederive LTD; we use the multiloop algorithmic construction of Ref. [1].

CFF dependency data. For CFF/causal representations we use the graph as the primary input and assume the canonical source/sink contraction ordering used by the implementation. Thus

$$I^{(3)}_{\mathrm{CFF}}[\mathcal{G};N](\vec{k};p,m) = \sum_{\mathfrak{o}\in\mathcal{O}_{\mathrm{CFF}}(\mathcal{G})} N\Big(Q^{\mathrm{CFF}}_{\mathfrak{o}}(\vec{k};p,m)\Big)\, W^{\mathrm{CFF}}_{\mathfrak{o}}(\vec{k};p,m). \tag{1.6}$$

The CFF weights use causal E-surfaces in the denominator. The construction is the one of Refs. [2, 3]. The standard black-box numerator assumption is that N is at most affine in each EMR edge energy q^0_e.

Surface notation. For any integer vector $c_e \in \{-1,0,1\}$ and external-energy shift Δ, define the signed causal surface

$$\eta(c,\Delta) = \sum_e c_e\,\omega_e + \Delta. \tag{1.7}$$

An E-surface has all non-zero internal-energy coefficients with the same orientation after the canonical CFF construction. An H-surface is a more general signed hyperboloid surface. Pure CFF denominator factors use only E-surfaces; LTD and hybrid terms may use H-surfaces when the residue geometry requires them. Numerator-side surfaces are allowed in the numerator and do not introduce denominator singularities.

1.2 Repeated Propagators Without Numerator Derivatives

Repeated propagators correspond to groups

$$R_g = \{r_{g,1},\ldots,r_{g,\nu_g}\}, \qquad \nu_g \geq 2, \tag{1.8}$$

whose members have the same canonical routing and mass, possibly with reversed edge orientation. Let Q^0_g be the canonical repeated-channel energy and ω_g the common on-shell energy. The purpose of the hybrid construction is to evaluate the repeated-channel residues without applying derivatives to the black-box numerator.

1.2.a The Fully Generic Formula

Start from the four-dimensional integral split into non-repeated edges S and repeated groups G_{rep}:

$$\begin{aligned} I_{\mathcal{G},\mathcal{B}}[N] &= \int \prod_{r=1}^{L} \frac{\mathrm{d}^4 k_r}{(2\pi)^4}\, N(\{q_e(k,p)\}) \prod_{s\in S} D_s^{-1} \prod_{g\in G_{\mathrm{rep}}} \prod_{a=1}^{\nu_g} D^{-1}_{r_{g,a}} \\ &= \int \prod_{r=1}^{L} \frac{\mathrm{d}^4 k_r}{(2\pi)^4}\, N(\{q_e(k,p)\}) \prod_{s\in S} D_s^{-1} \prod_{g\in G_{\mathrm{rep}}} D_g^{-\nu_g}. \end{aligned} \tag{1.9}$$

The second equality is only a denominator identity. The numerator still has one EMR argument for every original edge $e \in E$, including all repeated copies. Thus the collapsed channel g has a single denominator energy Q^0_g, but the numerator still sees $q^0_{r_{g,a}}$ for $a = 1,\ldots,\nu_g$.

Let $\bar{E} = S \sqcup G_{\mathrm{rep}}$ be the collapsed edge set and assign powers

$$n_{\bar{e}} = \begin{cases} 1, & \bar{e} = s \in S, \\ \nu_g, & \bar{e} = g \in G_{\mathrm{rep}}. \end{cases}$$

For every rank-L collapsed cut basis $B \subset \bar{E}$ and pole-sign vector σ_B, solve the cut equations

$$Q^0_b(k^0,p^0) = \sigma_b\omega_b, \qquad b \in B, \tag{1.10}$$

for the loop energies:

$$k^0 = K_{B,\sigma_B}(\vec{k}; p, m). \tag{1.11}$$

The Jacobian associated with this change of variables is

$$J_B = \det(\rho_{br})_{b\in B,\, r=1,\ldots,L}. \tag{1.12}$$

With the contour convention used in the implementation, the sign carried by the corresponding simple LTD residue is denoted by $s_{B,\sigma_B} \in \{\pm 1\}$. The direct residue theorem gives the strict equality

$$I_{\mathcal{G},\mathcal{B}}[N] = \int \prod_{r=1}^{L} \frac{\mathrm{d}^3 k_r}{(2\pi)^3} \sum_{B,\sigma_B} s_{B,\sigma_B} \Bigg[\prod_{b\in B} \frac{1}{(n_b-1)!} \left(\frac{\partial}{\partial Q_b^0}\right)^{n_b-1} \frac{N(\{q_e^0(k^0,p^0), \vec{q}_e\})}{\prod_{b\in B}(Q_b^0+\sigma_b\omega_b)^{n_b} \prod_{\bar{e}\notin B} D_{\bar{e}}^{n_{\bar{e}}}} \Bigg]_{k^0=K_{B,\sigma_B}}. \tag{1.13}$$

The determinant J_B is already included in the LTD residue sign and normalization used to define s_{B,σ_B}. For the DOT examples all non-zero J_B are ± 1. If a different integer LMB gives $|J_B| \neq 1$, the residue weight is multiplied by $1/|J_B|$.

Equation (1.13) is not yet acceptable when a powered channel is cut, because the derivatives act on N. Define the set of active original edges

$$\mathcal{A}_B = \left\{ e \in E \;:\; q_e^0 \text{ depends on } Q_b^0 \text{ for some } b \in B \text{ with } n_b > 1 \right\}. \tag{1.14}$$

For the affine EMR class, one exact algebraic route is to insert the multilinear pole interpolation identity

$$N(\{q_e^0\}) = \sum_{\theta\in\{\pm 1\}^{\mathcal{A}_B}} N\big(\{q_{e,\theta}^0\}\big) \prod_{e\in\mathcal{A}_B} \frac{1+\theta_e q_e^0/\omega_e}{2}, \tag{1.15}$$

with $q_{e,\theta}^0 = \theta_e\omega_e$ for active edges and the ordinary affine value for inactive edges. Since N is affine in each active EMR energy, differentiating (1.15) differentiates only the known interpolation factors and never the black-box function values.

The form actually used for the hybrid repeated sectors is the equivalent physical CFF-localized form. To derive it independently, temporarily split the repeated denominator into distinct on-shell energies $\omega_{g,a}$, $a = 1, \ldots, \nu_g$, and use the linear-pole decomposition

$$\frac{1}{(Q_g^0)^2 - \omega_{g,a}^2 + \mathrm{i}0} = \sum_{\tau_{g,a}=\pm 1} \frac{\tau_{g,a}}{2\omega_{g,a}} \frac{1}{Q_g^0 - \tau_{g,a}\omega_{g,a} + \mathrm{i}0\,\tau_{g,a}}. \tag{1.16}$$

Each linear pole has the causal exponential representation

$$\frac{1}{Q_g^0 - \tau\omega + \mathrm{i}0\,\tau} = -\mathrm{i}\,\tau \int_0^\infty \mathrm{d}\lambda\, \exp\!\big[\mathrm{i}\tau\lambda Q_g^0 - \mathrm{i}\lambda\omega\big]. \tag{1.17}$$

For a repeated group this gives a sign orbit $\tau_g = (\tau_{g,1}, \ldots, \tau_{g,\nu_g})$ and the phase

$$\Phi_{\tau_g} = \exp\left[\mathrm{i}Q_g^0 \sum_{a=1}^{\nu_g} \tau_{g,a}\lambda_{g,a} - \mathrm{i}\sum_{a=1}^{\nu_g} \lambda_{g,a}\omega_{g,a}\right], \qquad \lambda_{g,a} \geq 0. \tag{1.18}$$

The loop-energy integrations are now elementary Fourier integrations. For a complete set of CFF/LTD pole choices P, with signs τ_j, energies ω_j, and loop-energy rows ρ_j, one obtains the localizing constraint

$$\prod_{r=1}^{L} 2\pi\,\delta\left(\sum_{j\in P} \tau_j\lambda_j\rho_{jr}\right). \tag{1.19}$$

Solving (1.19) with $\lambda_j \geq 0$ is the standard CFF source/sink contraction step: the surviving sign assignments are precisely the causal orientations. In a repeated group the individual $\tau_{g,a}$ survive as numerator-sampling signs, while the localized collapsed chain has a single sign $\chi_g = \pm 1$ for the energy Q_g^0. If all repeated-copy signs are equal, χ_g is forced to that common sign. If the orbit is mixed, both $\chi_g = \pm 1$ are possible localizations. This is exactly the rule implemented as

$$\tau_g = (+,+,+) \Rightarrow \chi_g = +, \qquad \tau_g = (-,-,-) \Rightarrow \chi_g = -, \qquad \text{mixed } \tau_g \Rightarrow \chi_g = \pm.$$

Only after solving the delta constraints do we take the equal-energy limit $\omega_{g,a} \to \omega_g$. The split expression is an ordinary simple-pole CFF/LTD expression, hence the limit is equal to the confluent residue (1.13). The difference is organizational:

- the confluent derivative form differentiates the known denominator factors and would differentiate N unless (1.15) is inserted first;
- the physical CFF-localized form samples N directly at repeated-copy on-shell sign orbits (τ_g, χ_g), then combines those samples with rational affine weights.

For an affine EMR numerator the two forms are identical because both are the same equal-energy limit of the split simple-pole identity. They differ only in the basis of numerator calls: the direct interpolation basis uses the formal active edge signs θ_e, while the physical basis uses the CFF-localized orbits (τ_g, χ_g). The latter is the one printed by the current hybrid JSON labels such as `physG3tmmmcm`.

It remains to define the rational weights multiplying those physical samples. Let $\mathcal{X}_{B,\sigma}$ be the finite set of physical samples obtained from the above sign-orbit rule for the active repeated groups of the cut (B, σ_B). A sample $x \in \mathcal{X}_{B,\sigma}$ determines complete loop and edge energy maps,

$$K_x^0(\vec{k}; p, m), \qquad q_{e,x}^0(\vec{k}; p, m).$$

Write every component of every map as an affine vector in the symbols

$$1, \qquad p_a^0, \qquad \omega_{\bar{e}},$$

where repeated copies share the collapsed symbol ω_g. Let R_x be the column vector containing all those affine coefficients for sample x. For a numerator-derivative multi-index β, let T_β be the target affine vector:

- for $\beta = 0$, T_0 is the ordinary cut map $K^0_{B,\sigma_B}, q^0_{e,B,\sigma_B}$;
- for $|\beta| = 1$, T_β is the first derivative of those maps with respect to the corresponding repeated cut energy, multiplied by $2\omega_g$. The compensating factor $1/(2\omega_g)$ is stored as one additional repeated half-edge in the JSON.

The sample coefficients $\lambda_{\beta x}$ are defined by the affine system

$$\sum_{x \in \mathcal{X}_{B,\sigma}} \lambda_{\beta x} = \delta_{\beta 0}, \qquad \sum_{x \in \mathcal{X}_{B,\sigma}} \lambda_{\beta x} R_x = T_\beta. \tag{1.20}$$

If the system is underdetermined, the implementation uses the minimum Euclidean norm solution. Equivalently, after discarding dependent rows, if R is the matrix with columns R_x and t_β is the target vector including the constant row, then

$$\lambda_\beta = R^T (R R^T)^{-1} t_\beta. \tag{1.21}$$

This equation is not a fit or approximation: the affine support of $\mathcal{X}_{B,\sigma}$ is checked exactly over rational numbers.

Substituting (1.15) into (1.13) gives

$$I_{\mathcal{G},\mathcal{B}}[N] = \int \prod_{r=1}^{L} \frac{\mathrm{d}^3 k_r}{(2\pi)^3} \, I^{\mathrm{HybridLTD}}_{\mathcal{G},\mathcal{B},\kappa}[N](\vec{k};p,m), \tag{1.22}$$

$$I^{\mathrm{HybridLTD}}_{\mathcal{G},\mathcal{B},\kappa}[N] = \sum_{\mathfrak{o}\in\mathcal{O}_{\mathrm{H}}(\mathcal{G},\mathcal{B},\kappa)} N\Big(Q^{\mathrm{H}}_{\mathfrak{o}}(\vec{k};p,m)\Big)\, W^{\mathrm{H}}_{\mathfrak{o}}(\vec{k};p,m). \tag{1.23}$$

Here one orientation is

$$\mathfrak{o} = (B, \sigma_B, x, \beta, \gamma, \mu),$$

where $x \in \mathcal{X}_{B,\sigma}$, β is the part of the confluent derivative reconstructed from numerator samples, $\gamma = \alpha - \beta$ acts on known denominator factors, and μ labels one algebraic term in the known denominator derivative expansion. Here

$$\alpha_b = n_b - 1, \qquad b \in B.$$

The numerator is never differentiated; it is only sampled at

$$Q^{\mathrm{H}}_{\mathfrak{o}} = \{q^0_{e,x}, \vec{q}_e\}_{e\in E}.$$

The denominator derivative coefficient is also explicit. Each known factor is of the form

$$F_j(y) = \zeta_j(y)^{-p_j}, \qquad \zeta_j(y) = \zeta_j(0) + a_j \cdot y,$$

where $y = (Q^0_b - \sigma_b\omega_b)_{b\in B}$, p_j is either a propagator power or a surface power, and a_j is the vector of derivatives with respect to the cut-basis energies. A multi-derivative assignment δ_j with $\sum_j \delta_j = \gamma$ contributes

$$d_{\{\delta_j\}} = \frac{\gamma!}{\prod_j \delta_j!} \prod_j (-1)^{|\delta_j|} (p_j)_{|\delta_j|} a_j^{\delta_j}, \qquad (p)_r = p(p+1)\cdots(p+r-1). \tag{1.24}$$

For multi-indices, $\gamma! = \prod_b \gamma_b!$, $\delta_j! = \prod_b (\delta_j)_b!$, and

$$a_j^{\delta_j} = \prod_b (a_j)_b^{(\delta_j)_b}.$$

For a surface factor $\zeta_j = \eta_j$, the derivative raises the denominator power of that cached surface. For a cut half-edge factor $(2\omega_b)^{-p_j}$, it raises the number of stored half-edge entries.

The complete rational coefficient multiplying one JSON variant is therefore

$$c_{\mathfrak{o}} = s_{B,\sigma_B} \frac{1}{\alpha!} \binom{\alpha}{\beta} \lambda_{\beta x}\, d_\mu, \tag{1.25}$$

where d_μ is the value of (1.24) for the derivative assignment $\mu = \{\delta_j\}$, and

$$\alpha! = \prod_{b\in B} \alpha_b!, \qquad \binom{\alpha}{\beta} = \prod_{b\in B} \binom{\alpha_b}{\beta_b}.$$

The weight is the finite product

$$W^{\mathrm{H}}_{\mathfrak{o}} = c_{\mathfrak{o}} \prod_{h\in H_{\mathfrak{o}}} \frac{1}{2\omega_h} \prod_{\eta\in D_{\mathfrak{o}}} \eta(\vec{k};p,m)^{-m_{\mathfrak{o},\eta}} \prod_{\zeta\in N_{\mathfrak{o}}} \zeta(\vec{k};p,m). \tag{1.26}$$

Here $H_{\mathfrak{o}}$ is the multiset of half-edge ids, so each $h \in H_{\mathfrak{o}}$ contributes $(2\omega_h)^{-1}$. The denominator multiset $D_{\mathfrak{o}}$ contains cached surfaces

$$\eta(\vec{k};p,m) = \sum_e c_e\omega_e + \sum_a d_a p^0_a + c_0,$$

with $c_e, d_a, c_0 \in \mathbb{Z}$. The numerator multiset $N_{\mathfrak{o}}$ contains cached numerator-only surfaces of the same linear form; each $\zeta \in N_{\mathfrak{o}}$ is such a known linear factor in on-shell and external energies. Thus the previously used symbol $\zeta(\vec{k}; p, m)$ means exactly a cached linear surface factor appearing in the numerator, not an additional denominator. If no repeated channel is present, then $n_b = 1$ for all cuts, $\mathcal{A}_B = \varnothing$, and (1.23) collapses exactly to the LTD representation with the same $(\mathcal{G}, \mathcal{B}, \kappa)$.

1.2.b The Box with a Triple Repeated Propagator

Consider the one-loop box routing

$$q_0 = k, \qquad q_1 = k + p_1, \qquad q_2 = k + p_1 + p_2, \qquad Q = k + p_1 + p_2 + p_3,$$

with three repeated copies of Q. The collapsed denominator is

$$D_0 D_1 D_2 D_Q^3.$$

Write $\Delta_i = p_1^0 + \cdots + p_i^0$ and define the causal surfaces

$$E_{ij}^{\sigma_i \sigma_j} = \sigma_i \omega_i + \sigma_j \omega_j + \Delta_i - \Delta_j, \tag{1.27}$$

$$E_{iQ}^{\sigma_i \sigma_Q} = \sigma_i \omega_i + \sigma_Q \omega_Q + \Delta_i - \Delta_3. \tag{1.28}$$

The three simple-cut sectors are

$$I_i^{\text{simple}} = -\frac{N\left(q_0^0, q_1^0, q_2^0, Q^0, Q^0, Q^0\right)_{k^0 = -\Delta_i + \omega_i}}{2\omega_i \, E_{ij}^{++} \, E_{i\ell}^{++} \, \left(E_{iQ}^{++}\right)^3}, \qquad \{i, j, \ell\} = \{0, 1, 2\}. \tag{1.29}$$

The signs shown correspond to the upper on-shell pole convention used in the examples; the opposite closure is obtained by replacing every displayed $+\omega$ by the corresponding signed on-shell energy.

For the repeated cut used in the printed example, the collapsed LTD cut is $Q^0 = +\omega_Q$, hence

$$k^0 = \omega_Q - \Delta_3.$$

The six denominator surfaces generated by the three non-cut edges are

$$\eta_{18} = -\omega_0 + \omega_Q - \Delta_3, \qquad \eta_{19} = \omega_0 + \omega_Q - \Delta_3, \tag{1.30}$$

$$\eta_{20} = -\omega_1 + \omega_Q - \Delta_2, \qquad \eta_{21} = \omega_1 + \omega_Q - \Delta_2, \tag{1.31}$$

$$\eta_{22} = -\omega_2 + \omega_Q - \Delta_1, \qquad \eta_{23} = \omega_2 + \omega_Q - \Delta_1. \tag{1.32}$$

These are precisely surface ids $18, \ldots, 23$ in the CLI output. The overview table prints a compact subset of denominator ids, but `-show-details-for-orientation` displays the full factorization tree containing all six surfaces.

Let $\tau = (\tau_3, \tau_4, \tau_5)$ be the physical repeated-copy signs and χ the localized collapsed-chain sign. A physical numerator call is

$$N_{\tau,\chi} = N(\chi\omega_Q - \Delta_3, \ \chi\omega_Q - \Delta_2, \ \chi\omega_Q - \Delta_1, \ \tau_3\omega_Q, \ \tau_4\omega_Q, \ \tau_5\omega_Q). \tag{1.33}$$

The admissible physical sample set is

$$\mathcal{X}_Q = \{(+++,+), (---,-)\} \cup \{(\tau,+), (\tau,-) : \tau \in \{\pm\}^3, \ \tau \neq +++, \ \tau \neq ---\}. \tag{1.34}$$

For each $x = (\tau, \chi) \in \mathcal{X}_Q$, the affine coefficients λ_{0x} and λ_{1x} defined by (1.20) reconstruct respectively the numerator itself and the first numerator derivative. The full table is

sample $x = (\tau, \chi)$	λ_{0x}	λ_{1x}
$(+++,+)$	$43/112$	$5/8$
$(---,-)$	$-27/112$	$-5/8$
$(--+,-)$	$-3/56$	$-1/4$
$(--+,+)$	$1/112$	$-1/8$
$(-+-,-)$	$-3/56$	$-1/4$
$(-+-,+)$	$1/112$	$-1/8$
$(-++,-)$	$15/112$	$1/8$
$(-++,+)$	$11/56$	$1/4$
$(+--,-)$	$-3/56$	$-1/4$
$(+--,+)$	$1/112$	$-1/8$
$(+-+,-)$	$15/112$	$1/8$
$(+-+,+)$	$11/56$	$1/4$
$(++-,-)$	$15/112$	$1/8$
$(++-,+)$	$11/56$	$1/4$

The first column sums to one and reconstructs the base affine numerator map. The second column sums to zero and reconstructs the derivative map, with one extra factor $1/(2\omega_Q)$.

For the repeated box, $\alpha = 2$. Hence $\beta = 0, \gamma = 2$ or $\beta = 1, \gamma = 1$. Define

$$P = \prod_{i=18}^{23} \eta_i^{-1}, \qquad h_r = \sum_{18 \le i_1 \le \cdots \le i_r \le 23} \eta_{i_1}^{-1} \cdots \eta_{i_r}^{-1}, \qquad h_0 = 1.$$

The denominator derivative terms are

$$\gamma = 2: \quad 2\,\frac{Ph_2}{(2\omega_Q)^3} + 6\,\frac{Ph_1}{(2\omega_Q)^4} + 12\,\frac{P}{(2\omega_Q)^5}, \tag{1.35}$$

$$\gamma = 1: \quad -\frac{Ph_1}{(2\omega_Q)^3} - 3\,\frac{P}{(2\omega_Q)^4}. \tag{1.36}$$

These equations are exactly the two factorization trees printed in the JSON: Ph_2 is the large tree with all weakly ordered pairs of surfaces, Ph_1 is the smaller tree, and P is the single chain $[18, 19, 20, 21, 22, 23]$.

The complete repeated-cut contribution is therefore

$$I_Q^{\text{rep}} = \sum_{x \in \mathcal{X}_Q} N_x \left[-\frac{\lambda_{0x}}{2} \left(2\,\frac{Ph_2}{(2\omega_Q)^3} + 6\,\frac{Ph_1}{(2\omega_Q)^4} + 12\,\frac{P}{(2\omega_Q)^5} \right) - \lambda_{1x} \left(-\frac{Ph_1}{(2\omega_Q)^4} - 3\,\frac{P}{(2\omega_Q)^5} \right) \right]. \tag{1.37}$$

The overall minus sign is the one-loop LTD residue sign for the Q-cut in this routing. Equation (1.37) contains no undefined coefficient: every rational number is either one of the affine weights in the table or one of the derivative factors in (1.35)–(1.36).

For the concrete orientation requested in the CLI output,

$$x = (---,-), \qquad N_x = N_{---,-}.$$

The coefficients are

$$\lambda_{0x} = -\frac{27}{112}, \qquad \lambda_{1x} = -\frac{5}{8}.$$

Using (1.25), the $\beta = 0, \gamma = 2$ variants are

$$-\frac{1}{2}\lambda_{0x}\,(2, 6, 12) = \left(\frac{27}{112}, \frac{81}{112}, \frac{81}{56} \right),$$

with half-edge powers $3, 4, 5$. The $\beta = 1, \gamma = 1$ variants include the extra numerator-derivative half-edge and have

$$-\lambda_{1x}(-1, -3) = \left(-\frac{5}{8}, -\frac{15}{8}\right),$$

with half-edge powers $4, 5$. These five numbers are exactly

variant label	prefactor	half-edge multiset
`B3|b0|g2|n0:physG3tmmmcm`	$27/112$	$[3, 3, 3]$
`B3|b0|g2|n0:physG3tmmmcm`	$81/112$	$[3, 3, 3, 3]$
`B3|b0|g2|n0:physG3tmmmcm`	$81/56$	$[3, 3, 3, 3, 3]$
`B3|b1|g1|nD0phys:physG3tmmmcm`	$-5/8$	$[3, 3, 3, 3]$
`B3|b1|g1|nD0phys:physG3tmmmcm`	$-15/8$	$[3, 3, 3, 3, 3]$.

The label `tmmmcm` means $\tau = (-, -, -)$ and $\chi = -$; `b0/g2` means $\beta = 0, \gamma = 2$, and `b1/g1` means $\beta = 1, \gamma = 1$. The repeated edge signs in the orientation are all minus because the numerator sample uses the physical repeated-copy signs $\tau = (-, -, -)$, even though the denominator tree is the derivative expansion around the collapsed $Q^0 = +\omega_Q$ cut.

The final boxed formula is

$$I^{\text{HybridLTD}}_{\square(3)} = I_0^{\text{simple}} + I_1^{\text{simple}} + I_2^{\text{simple}} + I_Q^{\text{rep}}.$$

1.3 Higher Energy Powers in Numerators

The CFF construction is naturally compatible with numerators affine in each EMR edge energy. When a user supplies a bound d_e for the highest power of each edge energy in N, the implementation first checks loop-energy UV convergence. In every loop-energy direction v, the degree of the numerator minus twice the number of denominators depending on that direction must be strictly smaller than -1. If this check fails, finite pole data are not enough and the command refuses to build a representation.

1.3.1.a Quadratic Bounds

Let $x = q_e^0$, $\omega = \omega_e$, and suppose N has degree at most two in x, while all other EMR edge energies are held fixed. There is an exact polynomial division

$$N(x) = R_e N(x) + \left(x^2 - \omega^2\right) C_e N, \tag{1.38}$$

where the pole-safe affine remainder is

$$R_e N(x) = \frac{x + \omega}{2\omega} N(+\omega) + \frac{\omega - x}{2\omega} N(-\omega), \tag{1.39}$$

and the contact quotient is

$$C_e N = \frac{N(+\omega) - 2N(0) + N(-\omega)}{2\omega^2}. \tag{1.40}$$

Therefore

$$\frac{N}{D_e \prod_{f \neq e} D_f} = \frac{R_e N}{D_e \prod_{f \neq e} D_f} + \frac{C_e N}{\prod_{f \neq e} D_f}. \tag{1.41}$$

The first term is affine in q_e^0 and can be handled by ordinary CFF. The second term is a pinched graph with edge e removed, also handled by CFF. Consequently the pure CFF denominator remains E-surface only. The non-trivial part is to lift the CFF expression of the deleted graph back to an orientation of the original graph. The lift is completely mechanical:

1. Delete the pinched edge set P from the denominator graph and build a CFF representation of the lower graph $\mathcal{G} \setminus P$. If the lower graph factorizes into independent loop-energy components, build one CFF factor per component and multiply the component representations.

2. Every lower-sector CFF term fixes loop energies $K_{\mathrm{t}}^{0,\mathcal{G}\setminus P}$ in the original LMB. If component CFF factors were built separately, the global loop-energy map is reconstructed by solving the full-rank linear system defined by the union of component basis edges.

3. For every original edge, including each deleted edge, compute the lifted EMR energy

$$\widehat{q}^{\,0}_{e,\mathrm{t}} = \rho_e \cdot K_{\mathrm{t}}^{0,\mathcal{G}\setminus P} + \eta_e \cdot p^0. \tag{1.42}$$

   This number is used only for known polynomial factors. It is not automatically the numerator sample value of a deleted edge.

4. The numerator sample value of a deleted quadratic edge is one of $+\omega_e, 0, -\omega_e$, because $C_e N$ in (1.40) depends on those three black-box samples. The original JSON orientation therefore assigns $q_e^0 = +\omega_e$, 0, or $-\omega_e$ to the missing edge and records the corresponding orientation character $+$, 0, or $-$.

5. Any lifted factor $\widehat{q}^{\,0}_{e,\mathrm{t}}$ that remains outside the black-box numerator is stored as a numerator-side surface. In the strictly quadratic contact quotient no such q-factor is needed; for cubic and higher bounds it is needed.

6. The lower-sector CFF denominator surfaces are copied into the original surface cache after replacing every lower-sector edge id by its original edge id. No denominator surface involving a deleted edge is introduced.

The rational coefficients in the JSON follow directly from (1.40). Since half-edge lists encode $(2\omega_e)^{-1}$, the factor $1/(2\omega_e^2)$ is represented as

$$\frac{1}{2\omega_e^2} = \frac{2}{(2\omega_e)^2}.$$

The three samples $N(+\omega_e), N(0), N(-\omega_e)$ therefore carry raw half-edge prefactors $2, -4, 2$. A one-edge CFF pinch contributes one additional sign (-1) in the implementation's CFF orientation convention, so the printed pinched quadratic prefactors are $-2, +4, -2$.

In hybrid mode, the same division is applied to non-affine edge energies before or inside the repeated-channel residue expansion. For non-repeated edges this is identical to the pure CFF lift above. For repeated edges the division is combined with the repeated-channel sample set $\mathcal{X}_{B,\sigma}$. The black-box numerator is still sampled only at the explicit finite set of energies supplied by the JSON orientation map.

1.3.1.b One-Loop Box with One Quadratic Edge

For a non-repeated one-loop box with denominators $D_0D_1D_2D_3$, assume the only non-affine dependence is $d_0 = 2$. Use the routing

$$q_0 = k, \quad q_1 = k + p_1, \quad q_2 = k + p_1 + p_2, \quad q_3 = k + p_1 + p_2 + p_3,$$

and set $\Delta_i = p_1^0 + \cdots + p_i^0$. Equations (1.38)–(1.41) give

$$I_{\square}^{\mathrm{CFF}}[N] = I_{\mathrm{CFF}}[\square; R_0 N] + I_{\mathrm{CFF}}[\square/0; C_0 N], \tag{1.43}$$

where $\Box/0$ is the triangle obtained by deleting propagator 0. More explicitly,

$$I_{\mathrm{CFF}}[\Box; R_0 N] = \sum_{\mathfrak{o}\in\mathcal{O}_{\mathrm{CFF}}(\Box)} \left[\frac{q_0^0(\mathfrak{o})+\omega_0}{2\omega_0} N(q_0^0=+\omega_0) + \frac{\omega_0 - q_0^0(\mathfrak{o})}{2\omega_0} N(q_0^0=-\omega_0)\right] W_{\mathfrak{o}}^{\mathrm{CFF}}, \tag{1.44}$$

$$I_{\mathrm{CFF}}[\Box/0; C_0 N] = \sum_{\mathfrak{t}\in\mathcal{O}_{\mathrm{CFF}}(\Box/0)} \frac{N(+\omega_0) - 2N(0) + N(-\omega_0)}{2\omega_0^2} W_{\mathfrak{t}}^{\mathrm{CFF}}. \tag{1.45}$$

Both $W_{\mathfrak{o}}^{\mathrm{CFF}}$ and $W_{\mathfrak{t}}^{\mathrm{CFF}}$ contain only E-surface denominators.

The lifted triangle terms make the orientation structure concrete. If the triangle CFF term cuts edge $j \in \{1, 2, 3\}$ with sign σ_j, then

$$k^0 = \sigma_j \omega_j - \Delta_j, \qquad \widehat{q}_0^0 = \sigma_j \omega_j - \Delta_j.$$

However, in the contact numerator call the deleted edge energy is not $\widehat{q}_0^0$; it is one of the three interpolation samples

$$q_0^0 = +\omega_0, \qquad q_0^0 = 0, \qquad q_0^0 = -\omega_0.$$

For example, the lower triangle orientation with original signs $(-++)$ on edges $1, 2, 3$ contributes three lifted original-box orientations

$$+-++, \qquad 0-++, \qquad --++,$$

where the first character is the assigned numerator value of the deleted edge. The printed coefficients are

orientation	variant	prefactor	half-edge multiset
$+-++$	`pinch[0]|e0=+`	-2	$[0, 0, 1, 2, 3]$
$0-++$	`pinch[0]|e0=0`	4	$[0, 0, 1, 2, 3]$
$--++$	`pinch[0]|e0=-`	-2	$[0, 0, 1, 2, 3]$.

The denominator surfaces of that lower triangle are the ordinary CFF surfaces among edges $1, 2, 3$; in the CLI table they appear, for this orientation, as ids $[4, 5]$. The two extra half-edge entries $0, 0$ encode the factor $(2\omega_0)^{-2}$. No denominator singularity associated with edge 0 remains because D_0 has been cancelled by the contact quotient. This is why the final JSON is still a normal orientation of the original box: all four numerator edge energies are specified, but only the lower triangle denominator surfaces are present.

1.3.2.a Bounds Above Quadratic

For $d_e > 2$, the same idea works but the contact quotient is no longer a constant in the deleted edge energy. Let $a_1, \ldots, a_{d_e-1}$ be non-pole interpolation nodes with $a_j \neq \pm 1$, and write

$$u = \frac{q_e^0}{\omega_e}.$$

The implementation uses the deterministic node list

$$0,\ 2,\ -2,\ 3,\ -3, \ldots$$

truncated to $d_e - 1$ nodes. The Lagrange basis polynomial associated with node a_j is

$$L_j(u) = \prod_{\ell\neq j} \frac{u - a_\ell}{a_j - a_\ell}. \tag{1.46}$$

It is characterized by $L_j(a_i) = \delta_{ij}$. The pole remainder is still the affine interpolation at $u = \pm 1$, and the contact quotient is reconstructed from black-box samples by

$$C_e N(q_e^0) = \sum_j \frac{L_j(q_e^0/\omega_e)}{(a_j^2 - 1)\omega_e^2} \left[N(a_j \omega_e) - \frac{a_j + 1}{2} N(+\omega_e) - \frac{1 - a_j}{2} N(-\omega_e) \right]. \tag{1.47}$$

The quotient $C_e N$ is now a known polynomial in the lifted deleted-edge energy $\widehat{q}_{e,\mathrm{t}}^0$ of (1.42). Expanding each $L_j(u)$ gives

$$L_j(u) = \sum_{r=0}^{d_e - 2} \ell_{jr} u^r.$$

A monomial contribution is therefore

$$\ell_{jr} \frac{(\widehat{q}_{e,\mathrm{t}}^0)^r}{(a_j^2 - 1)\omega_e^{r+2}} \left[N(a_j \omega_e) - \frac{a_j + 1}{2} N(+\omega_e) - \frac{1 - a_j}{2} N(-\omega_e) \right]. \tag{1.48}$$

In JSON this becomes:

- $r + 2$ entries of edge e in `half_edges`, because $\omega_e^{-(r+2)} = 2^{r+2}(2\omega_e)^{-(r+2)}$;
- r numerator-side copies of the lifted surface $\widehat{q}_{e,\mathrm{t}}^0$, unless that linear form is zero;
- one numerator map with $q_e^0 = a_j \omega_e$, $+\omega_e$, or $-\omega_e$, depending on the black-box sample in the bracket.

For several higher-power edges, (1.48) suggests a recursive construction: pick an active edge, split its contribution into the pole remainder and the contact sector, then repeat the operation on the remaining bounded edges of the same graph or lower graph. Once a cubic or higher contact has been taken, the lower graph carries known polynomial factors

$$\prod_s (\widehat{q}_{e_s,\mathrm{t}}^0)^{r_s}.$$

If a second non-affine edge is then contacted, those known factors and the black-box numerator samples no longer separate into independent edge-wise finite-pole divisions. The completion is obtained by reducing those known polynomial factors against the remaining quadratic denominators, as described in Sec. 1.3.2.c below. Algebraically this is the lower-sector contact/infinity completion of the finite-pole recursion. In JSON it is still encoded with the same ingredients: ordinary E-surface denominator trees, half-edge factors, numerator-only linear factors, and one black-box numerator map per orientation. The only grammar extension is that a denominator tree may now contain a unit node, represented by a node with no surfaces and no children. Such a node evaluates to 1 and represents a terminal tadpole or unit lower-sector contact.

The implementation exposes the following pure-CFF higher-power cases:

1. arbitrary quadratic-or-lower caps $d_e \leq 2$, on one-loop, multiloop, split-mass, and repeated-signature graphs, provided the loop-energy UV check is satisfied;
2. arbitrary UV-convergent higher caps, including caps placed on repeated-signature edges, whenever every lower contact sector can be decomposed into non-zero loop-energy matroid components.

In these cases every pure-CFF denominator tree is built only from E-surfaces. Numerator-side cached factors may be E- or H-type linear forms, but the JSON marks them as numerator-only and the evaluator multiplies them in the numerator.

Pure CFF does not need a separate repeated-pole derivative construction: repeated propagators are repeated denominator factors in the CFF contour integral. Therefore the same lower-contact completion applies when the higher-power numerator cap is placed on one of several repeated-signature copies. The hybrid formula remains useful for the LTD-like repeated-channel organization, but it is not needed to make the pure-CFF high-power lift well-defined.

Hybrid mode has a different status. The hybrid repeated-channel formula is not constrained to keep pure-CFF denominator trees E-only: H-surfaces are allowed because the non-repeated part is LTD-like and CFF is used only to localize repeated channels. With `-energy-degree-bounds`, numerator derivatives in the repeated-channel residue are replaced by tensor-product finite-difference samples in the repeated basis coordinates. For a derivative multiindex β, the degree used along basis direction b is

$$D_b = \sum_e d_e \, \mathbf{1}\{\partial q_e^0/\partial z_b \neq 0\},$$

where z_b is the repeated basis energy. The finite-difference stencil is the node set $0, 1, -1, 2, -2, \ldots$ truncated to $D_b + 1$ nodes, with weights fixed by exact polynomial interpolation. Therefore the numerator is still a black box: the JSON only asks for samples at explicit EMR energy maps.

The implemented hybrid path is validated for arbitrary user-provided caps subject to the same loop-energy UV check and practical expression size. The test suite covers cubic, quartic, and quintic caps on repeated box, sunrise, kite, and iterated-bubble examples by comparing the hybrid value to the split-mass LTD limiting proxy. In the absence of repeated propagators, hybrid collapses exactly to LTD for any accepted bound vector.

1.3.2.b One-Loop Box with One Cubic Edge

For the same one-loop box, take $d_0 = 3$. The implementation chooses the two non-pole nodes

$$a_0 = 0, \qquad a_1 = 2.$$

The Lagrange basis is

$$L_0(u) = 1 - \frac{u}{2}, \qquad L_1(u) = \frac{u}{2}. \tag{1.49}$$

Substituting these two polynomials into (1.47) gives

$$\begin{aligned} C_0 N(q_0^0) = &-\frac{1 - u/2}{\omega_0^2}\left[N(0) - \frac{1}{2}N(+\omega_0) - \frac{1}{2}N(-\omega_0)\right] \\ &+ \frac{u/2}{3\omega_0^2}\left[N(2\omega_0) - \frac{3}{2}N(+\omega_0) + \frac{1}{2}N(-\omega_0)\right], \qquad u = \frac{\widehat{q}_0^0}{\omega_0}. \end{aligned} \tag{1.50}$$

For a lower triangle orientation $\mathfrak{t}$, the lifted deleted energy is

$$\widehat{q}_{0,\mathfrak{t}}^0 = \rho_0 \cdot K_{\mathfrak{t}}^{0,\square/0}.$$

For instance, if the triangle cut is edge 1 with sign $-$, then

$$\widehat{q}_{0,\mathfrak{t}}^0 = -\omega_1 - p_1^0,$$

which appears in the surface cache as a numerator-only E-surface. If the cut is edge 1 with sign $+$, the lifted surface is $\omega_1 - p_1^0$. These are the numerator-only surfaces printed as (e) in the cubic box output.

The monomial coefficients of (1.50) map directly to JSON variants. Before the one-edge CFF pinch sign, the finite-pole contact components for edge 0 are

sample q_0^0	prefactor	half-edge multiset	numerator surface
0	-4	$[0,0]$	$\varnothing$
0	4	$[0,0,0]$	$[\widehat{q}_0^0]$
$+\omega_0$	2	$[0,0]$	$\varnothing$
$+\omega_0$	-4	$[0,0,0]$	$[\widehat{q}_0^0]$
$-\omega_0$	2	$[0,0]$	$\varnothing$
$-\omega_0$	$-4/3$	$[0,0,0]$	$[\widehat{q}_0^0]$
$2\omega_0$	$4/3$	$[0,0,0]$	$[\widehat{q}_0^0]$.

The printed one-loop box CFF table includes the additional pinch sign, so the visible variants are

orientation label	variant	prefactor	meaning
$0-++$	`pinch[0]|e0=0`	4	$-(-4)\,N(q_0^0=0)/(2\omega_0)^2$
$0-++$	`pinch[0]|e0=0`	-4	$-(4)\,\widehat{q}_0^0\,N(q_0^0=0)/(2\omega_0)^3$
$x-++$	`pinch[0]|e0=2`	$-4/3$	$-(4/3)\,\widehat{q}_0^0\,N(q_0^0=2\omega_0)/(2\omega_0)^3$
$+-++$	`pinch[0]|e0=+`	-2	$-(2)\,N(q_0^0=+\omega_0)/(2\omega_0)^2$
$+-++$	`pinch[0]|e0=+`	4	$-(-4)\,\widehat{q}_0^0\,N(q_0^0=+\omega_0)/(2\omega_0)^3$
$--++$	`pinch[0]|e0=-`	-2	$-(2)\,N(q_0^0=-\omega_0)/(2\omega_0)^2$
$--++$	`pinch[0]|e0=-`	$4/3$	$-(-4/3)\,\widehat{q}_0^0\,N(q_0^0=-\omega_0)/(2\omega_0)^3$.

The label x in $x-++$ indicates that the deleted edge is sampled at the non-standard finite node $q_0^0 = 2\omega_0$, so its detailed energy map must be read from the orientation details rather than from a single $+$, $-$, or 0 symbol. All denominator surfaces in these rows are the lifted triangle CFF surfaces; $\widehat{q}_0^0$ is a numerator-only surface and cannot create a denominator singularity.

This cubic box example is the first member of the higher-contact recursion. If the other box edges are affine, for example $(d_0, d_1, d_2, d_3) = (3, 1, 1, 1)$, the formula above is exact and agrees with LTD. If one spectator is quadratic, for example $(3, 1, 2, 0)$, a second contact produces the terminal tadpole derived below. The implemented command now emits that completion as an explicit unit denominator node, so the resulting pure-CFF JSON remains E-surface-only in the denominator and agrees with LTD for all UV-convergent one-loop box monomial caps through the tested boundary $\sum_e d_e \le 6$. The same higher cap is allowed on a graph that contains repeated propagators, including when the capped edge is one of the repeated copies, because pure CFF treats those copies as ordinary repeated denominator factors.

1.3.2.c What the First Rejected Contact Sector Contains

This subsection records the diagnostic calculation behind the phrase "lower-sector infinity/contact completion". The calculation is now encoded in the pure-CFF builder, but it is useful because the first formerly rejected sector is simple enough to analyze completely.

Consider the one-loop box with edge energies

$$q_e^0 = k^0 + \Delta_e, \qquad D_e = (q_e^0)^2 - \omega_e^2, \qquad e = 0, 1, 2, 3,$$

where Δ_e is the external-energy shift of edge e and $\omega_e = E_e(\vec{k})$. Define

$$\delta_{ab} = \Delta_a - \Delta_b, \qquad q_a^0 = q_b^0 + \delta_{ab}.$$

The first sector that exposed the missing terminal branch was

$$(d_0, d_1, d_2, d_3) = (0, 0, 3, 3), \qquad N(q) = (q_2^0)^3 (q_3^0)^3.$$

For a cubic edge the finite-pole/contact division is just

$$(q_e^0)^3 = \omega_e^2 q_e^0 + D_e\, q_e^0. \tag{1.51}$$

The first term is the pole remainder and is handled by the finite-pole CFF samples. The second term cancels D_e and produces a contact lower sector. Taking the contact part on both cubic edges produces, among other already handled terms, the lower bubble

$$\frac{q_2^0 q_3^0}{D_0 D_1}. \tag{1.52}$$

This bubble is the first place where the naive recursion was incomplete. Writing the numerator in terms of the remaining edge 0,

$$q_2^0 = q_0^0 + \delta_{20}, \qquad q_3^0 = q_0^0 + \delta_{30},$$

gives the exact identity

$$\begin{aligned} q_2^0 q_3^0 &= (q_0^0)^2 + (\delta_{20} + \delta_{30}) q_0^0 + \delta_{20}\delta_{30} \\ &= D_0 + \omega_0^2 + (\delta_{20} + \delta_{30}) q_0^0 + \delta_{20}\delta_{30}. \end{aligned} \tag{1.53}$$

Therefore

$$\frac{q_2^0 q_3^0}{D_0 D_1} = \frac{\omega_0^2 + (\delta_{20} + \delta_{30}) q_0^0 + \delta_{20}\delta_{30}}{D_0 D_1} + \frac{1}{D_1}. \tag{1.54}$$

The first term has an affine numerator in the lower bubble and is a standard CFF object. The second term is a terminal lower-sector tadpole. It has no ordinary E-surface denominator left after the half-edge factor of edge 1; in the JSON grammar it would be represented as a unit denominator tree with the half-edge list [1].

With the close-below convention used by the code, this terminal contribution has the sign of the corresponding CFF branch and evaluates to

$$-\frac{1}{2\omega_1}$$

for the sector in (1.52). Direct numerical inspection of the hidden experimental recursion confirms exactly this. Before the terminal tree is restored, the experimental expression differs from LTD by $+1/(2\omega_1)$. For the same random kinematics used in the tests,

$$\omega_1 = 0.788665091489179\ldots, \qquad \frac{1}{2\omega_1} = 0.633982669444499\ldots.$$

The formerly rejected box sectors

$$(0,3,0,3), \quad (0,3,3,0), \quad (2,1,0,3), \quad (3,1,2,0),$$

and their permutations show the same pattern: the missing piece is a single terminal tadpole $\pm 1/(2\omega_i)$, with i determined by the last remaining denominator in the lower sector. In the intermediate recursion these branches are terms with no E-surface denominator chain. The schema now serializes them as a unit denominator node

`{surfaces: [], children: []}`,

so the evaluator multiplies by the half-edge factors and by one. With this unit-node encoding, all UV-convergent one-loop box monomial caps through the boundary $\sum_e d_e \le 6$ agree with LTD at arbitrary precision.

The natural one-loop generalization is consequently not another finite-pole sample, but a lower-sector normal-form recursion. Let S be the set of denominators left after some contacts and let $P(q^0)$ be the known polynomial factor multiplying the black-box numerator sample. Choose $a \in S$ and divide P by

$$D_a = (q_a^0)^2 - \omega_a^2$$

after rewriting all q_e^0 in terms of q_a^0:

$$P(q^0) = D_a\, Q_a(q^0) + R_a(q^0), \qquad \deg_{q_a^0} R_a \leq 1. \tag{1.55}$$

Then

$$\frac{P}{\prod_{e\in S} D_e} = \frac{R_a}{D_a \prod_{e\in S\setminus\{a\}} D_e} + \frac{Q_a}{\prod_{e\in S\setminus\{a\}} D_e}. \tag{1.56}$$

The first term is again a finite-pole CFF term because its numerator is affine in the active energy. The second term is the contact completion and is recursed on the smaller denominator set. When one denominator remains, (1.56) terminates in a tadpole; when no denominator remains, UV convergence of the full sector must make the sum of such unit polynomial branches cancel across contact histories.

The same mechanism extends to multiloop contact sectors whenever the known polynomial factors reduce to the span of the denominator components that remain. The smallest diagnostic graph is the two-loop penta-denominator

$$D_0(k_1 + k_2 + p_3)D_1(k_1)D_2(k_1 + p_1)D_3(k_2)D_4(k_2 + p_2).$$

For the UV-convergent bounds

$$(d_0, d_1, d_2, d_3, d_4) = (1, 0, 3, 0, 3),$$

one contact history sets $q_2^0 = q_4^0 = q_0^0 = q_1^0 = 0$ and leaves only

$$\frac{(p_1^0)(p_2^0 - p_3^0)}{D_3(k_2)}.$$

The original graph has two loop energies, but this lower sector has rank one. It is nevertheless not ambiguous: all dependence on the unlocalized k_1^0 has disappeared into external-energy factors. The completion is therefore a single terminal tadpole,

$$\frac{p_1^0(p_2^0 - p_3^0)}{2\omega_3},$$

times the half-edge factors accumulated along the contact path. The code handles this by decomposing the remaining denominator rows into vector-matroid components, building a CFF or terminal LTD factor for each non-zero component, and solving only the localized rank directions. Free loop-energy directions are assigned the particular value zero in the derived `loop_q0` map; the EMR `edge_q0` map is the numerator map and is unchanged by that bookkeeping choice.

This gives a practical generalization of (1.55)–(1.56): recurse on active higher-power or known-polynomial edges, reduce quotient terms against the remaining denominators, decompose the lower denominator into matroid components, and serialize terminal components as unit denominator nodes. The pure-CFF tests now include the one-loop box sectors above, repeated-signature box sectors with quartic caps on repeated copies, and the two-loop rank-deficient penta sectors; in every accepted case the denominator surface set is E-only and the value agrees with LTD or, for repeated propagators, with the hybrid value and the split-mass LTD limiting proxy.

Chapter 2

Python Implementation and JSON Format

2.1 Module Structure

The implementation is organized around an evaluator-complete intermediate format.

- `graph_io.py` parses DOT graphs, extracts momentum signatures, detects repeated channels, and builds split-mass reference graphs.
- `graph_signatures.py` implements the `graph_from_signatures` command, which turns products such as `prop(k1+p1,m)` into DOT graphs.
- `orientation_bundle.py` constructs LTD, CFF, hybrid, and bounded-degree contact bundles. This is where repeated channels are collapsed, active numerator maps are enumerated, and CFF lower minors are generated.
- `structure.py` serializes bundles to schema-v6 JSON and contains the arbitrary-precision Python evaluator.
- `pretty.py` renders the JSON in a human-readable form.
- `symbolica_eval.py` compiles a fixed-numerator JSON expression to Symbolica eager and assembly-backed evaluators.

2.2 Bounded-Degree Support Matrix

The command-line option `-energy-degree-bounds` is interpreted as an upper bound d_e for the degree of the black-box numerator in each EMR edge energy q_e^0. The builder first applies the loop-energy UV check described in Section 1.3. If the check fails, no representation is built.

After the UV check, the guaranteed support is:

family	graph class	supported bounds
`ltd`	any graph	any UV-convergent bounds; structure unchanged
`hybrid`	no repeated propagators	any UV-convergent bounds; exact LTD collapse
`hybrid`	repeated propagators	any tested UV-convergent bounds; finite-difference samples
`cff`	any graph	all $d_e \leq 2$, denominator E-surfaces only
`cff`	high-power edge caps	UV-convergent caps with lower-contact completion; denominator E-surfa

The last `cff` row includes one-loop and multiloop sectors with mixed cubic/quadratic and quartic-or-higher caps, including caps on repeated-signature edges, provided the loop-energy UV check succeeds and the lower-contact completion can be decomposed into non-zero loop-energy matroid components.

The hybrid high-power tests are not pure-CFF tests. They compare the hybrid value with a split-mass LTD proxy and verify convergence as the mass split is reduced. This is the correct reference for repeated propagators because ordinary LTD on the unsplit repeated graph would require numerator derivatives, which the hybrid construction avoids.

2.3 Oriented JSON Semantics

The top-level JSON contains graph metadata, a surface cache, and a list of orientations. The central invariant is:

> one orientation corresponds to one unique EMR edge-energy numerator map.

All denominator terms sharing that numerator call are stored as `variants`. Each variant has its own rational prefactor, half-edge list, numerator-surface list, and factorized denominator tree.

The evaluator uses only the JSON plus the DOT graph. It does not reconstruct a hidden LTD or CFF object. A variant contributes

$$\texttt{pref} \prod_{e\in\texttt{half\_edges}} (2\omega_e)^{-1} \prod_{s\in\texttt{num\_surfaces}} \eta_s \prod_{t\in\texttt{denominator tree}} \eta_t^{-1}$$

times the numerator sampled at the orientation's `edge_q0` map.

2.4 Surface Cache

Each surface cache entry records a class, coefficients of internal on-shell energies, and external-energy shifts. The class is printed as `e` or `h`. If a surface appears only in numerator factors, the entry carries `numerator_only=true` and pretty printing displays `(e)` or `(h)`. If a numerator-side surface is identical to an existing denominator surface, the same id is reused.

2.5 Orientation Labels and Variants

Compact labels use $+$, $-$, 0, and x:

- $+$: the numerator samples that edge at $+\omega_e$;
- $-$: the numerator samples that edge at $-\omega_e$;
- 0: the numerator samples that edge at zero energy;
- x: the edge has a non-trivial affine map, printed by `-show-details`.

The physical origin of a contribution, such as CFF, LTD, pinched contacts, or hybrid basis/interpolation data, is variant metadata rather than part of the orientation identity. This is what keeps the number of numerator calls minimal.

2.6 Command-Line Workflow

The main commands are:

```
python3 hybrid3d.py validate --dot examples/graphs/box_pow3.dot
python3 hybrid3d.py build --family hybrid --dot examples/graphs/box_pow3.dot --pretty
python3 hybrid3d.py test --dot examples/graphs/box_pow3.dot
python3 hybrid3d.py test-cff-ltd --dot examples/graphs/box.dot --energy-degree-bounds
    0:2
python3 hybrid3d.py compile --orientation-json demo.json --dot graph.dot --numerator-
    expr "1"
python3 hybrid3d.py evaluate --orientation-json demo.json --dot graph.dot --evaluator-
    backend symbolica
```

The repeated-propagator three-way test compares pure CFF, hybrid, and the split-mass LTD limiting proxy. The ordinary LTD formula on an unsplit repeated graph is not a valid physics comparison because numerator derivatives are not included there.

2.7 Symbolica Evaluators

The `compile` command accepts one JSON representation and one fixed numerator expression. It reconstructs the JSON expression symbolically, introducing structured functions such as `OSE(qx,qy,qz,m)`, `etaE(i)`, `etaH(i)`, and tagged edge/loop/external momentum components. The resulting Symbolica evaluator is saved both as a shared library and as eager evaluator state. The JSON is updated with relative paths and fingerprints tying the evaluator to the graph, numerator, value type, and orientation structure.

The `evaluate` command can use:

`builtin`, `symbolica_compiled`, `symbolica_eager`, `symbolica_eager_symjit`.

Profiling performs ten calls of a batch evaluator. The default batch size is 100, so a profile line reports one thousand repeated samples of the same input kinematics.

The same saved eager evaluator is used by `evaluate -stability`. This mode converts the generated or supplied real kinematic input to decimal precision-tracking numbers with 16 significant digits by default, evaluates with Symbolica's `Evaluator.evaluate_with_prec`, and reports the number of significant digits carried by the final decimal result. The work precision for constants and arithmetic defaults to 80 decimal digits and can be changed with `-stability-work-precision`. Non-finite outputs are assigned zero reported digits. The printed central value is diagnostic only: once the reported digit count is small, the value may have lost enough significance to be numerically meaningless.

Chapter 3

Performance

3.1 Benchmark Setup

The numbers below were generated on April 26, 2026 using the scripts in `examples/scripts`. Each timing is the per-sample time reported by `evaluate -profiling 100`, i.e. ten batch calls of size 100. The 15-external one-loop benchmark is intentionally omitted here: the LTD build is small, but the pure CFF one-loop representation has $2^{15} - 2 = 32766$ sign sectors and the CFF build is not practical for routine documentation runs.

3.2 Repeated Box

The repeated box script `examples/scripts/box_pow3_cli_showcase.sh` compares hybrid and pure CFF for the six-edge graph with a triple repeated propagator. The numerator is `dot(edges[0], ext[0]) + edges[3][0]`.

label	backend	orientations	maps	value type	per sample
hybrid	compiled	17	17	real	0.0955 μs
hybrid	eager	17	17	real	0.521 μs
hybrid	eager+symjit	17	17	real	0.104 μs
hybrid	compiled	17	17	complex	0.271 μs
hybrid	eager	17	17	complex	0.725 μs
hybrid	eager+symjit	17	17	complex	0.0989 μs
CFF	compiled	62	62	real	0.162 μs
CFF	eager	62	62	real	1.44 μs
CFF	eager+symjit	62	62	real	0.142 μs
CFF	compiled	62	62	complex	0.466 μs
CFF	eager	62	62	complex	2.03 μs
CFF	eager+symjit	62	62	complex	0.136 μs

The hybrid representation has fewer numerator maps than CFF, which explains the faster compiled real evaluator. Complex compiled evaluation is slower because the generated function propagates complex arithmetic even though the input kinematics are real. Symjit eager evaluation is competitive for this small expression because the batch is small and the expression has only 17 or 62 maps.

3.3 One-Loop 10-Gon

The script `examples/scripts/one_loop_10_external_runtime_compare.sh` compares numerator-free LTD and CFF on a true one-loop 10-gon.

family	backend	orientations	maps	per sample
LTD	compiled	10	10	$0.234\,\mu s$
LTD	eager	10	10	$0.852\,\mu s$
LTD	eager+symjit	10	10	$0.147\,\mu s$
CFF	compiled	1022	1022	$3.96\,\mu s$
CFF	eager	1022	1022	$36.6\,\mu s$
CFF	eager+symjit	1022	1022	$2.97\,\mu s$

The result is expected: the one-loop CFF orientation count grows like $2^n - 2$, while LTD has only one cut per internal edge. The CFF expression is still fast in absolute terms after compilation, but it is much larger.

3.4 Five-Loop Four-External Benchmarks

The script `examples/scripts/five_loop_symbolica_runtime_compare.sh` compares three numerator choices on a non-repeated five-loop graph and on the repeated five-loop graph. The three numerator choices are 1, one edge-external dot product per internal edge, and the same numerator with three selected edge terms squared.

case	family	maps	compiled	eager	eager+symjit
no repeats, $N = 1$	LTD	137	0.326	1.88	0.250
no repeats, $N = 1$	CFF	930	1.26	12.3	0.892
no repeats, $N = 1$	hybrid	137	0.333	1.87	0.246
no repeats, linear	LTD	137	0.481	3.29	0.379
no repeats, linear	CFF	930	1.96	17.5	1.35
no repeats, linear	hybrid	137	0.484	3.24	0.391
no repeats, squared	LTD	137	0.493	3.45	1.73
no repeats, squared	CFF	4692	18.7	45.9	13.2
no repeats, squared	hybrid	137	0.520	3.31	0.390

case	family	maps	compiled	eager	eager+symjit
repeated, $N = 1$	LTD	137	0.334	1.90	0.272
repeated, $N = 1$	CFF	930	1.26	12.5	0.899
repeated, $N = 1$	hybrid	630	0.823	4.88	0.579
repeated, linear	LTD	137	0.481	3.44	0.408
repeated, linear	CFF	930	1.88	17.4	1.32
repeated, linear	hybrid	630	21.4	37.8	12.4
repeated, squared	LTD	137	0.491	3.53	0.387
repeated, squared	CFF	see text	see text	see text	see text
repeated, squared	hybrid	115	1.84	9.47	1.11

For non-repeated graphs the hybrid family collapses to the LTD structure, so their orientation and map counts agree. CFF is slower because it has more causal sign sectors. The squared CFF no-repeat case is much larger because the bounded-degree contact completion adds pinched CFF minors.

For repeated graphs, the LTD rows are included only as runtime references. The ordinary unsplit LTD expression does not include numerator derivatives for raised propagators; the correct comparison is CFF versus hybrid versus the split-mass LTD limiting test. The repeated unit numerator shows the expected middle ground: hybrid has fewer maps than CFF but more than

LTD. The repeated linear case is slower for hybrid because the confluent interpolation creates heavier denominator variants per numerator map. In the squared repeated case, the pure CFF row now uses the recursive E-surface-only contact/minor construction rather than the older hybrid fallback; the timing table should be regenerated before quoting a stable number for that row. The bounded hybrid reduction has 115 maps for this benchmark.

3.5 Precision Stability

The following tables use `evaluate -stability` with 16 significant digits in the input and 80 decimal digits of work precision. They measure the precision carried by Symbolica's eager precision-tracking arithmetic through the already compiled structured expression. These measurements are not identity tests; they probe numerical conditioning of the particular representation and kinematic point. When only one or two digits remain, the displayed value should be interpreted as a symptom of cancellation rather than as a trustworthy estimate.

example	family	maps	result digits	tracked value
repeated box	LTD	6	0	∞
repeated box	CFF	62	1	10^{-2}
repeated box	hybrid	17	1	-2×10^3
one-loop 10-gon	LTD	10	2	2.6×10^5
one-loop 10-gon	CFF	1022	1	10^{-1}

For the repeated box, the LTD row is again not a valid repeated-propagator formula; its singular output is a useful reminder that the unsplit LTD structure contains zero repeated-channel surfaces. CFF and hybrid have the same ordinary double value for this point, but the precision-tracking run shows that both are badly conditioned at 16 input digits, with the hybrid central value already unusable. This is consistent with the hybrid expression being a more compact but more cancellation-prone confluent combination.

five-loop case	LTD digits	CFF digits	hybrid digits
no repeats, $N = 1$	2	2	2
no repeats, linear	2	2	2
no repeats, squared	2	2	2
repeated, $N = 1$	0	2	2
repeated, linear	0	2	2
repeated, squared	0	2	2

The five-loop tables show that the tested random point is numerically delicate for all structured sums at ordinary double-like input precision. The non-repeated hybrid rows coincide with LTD, as required by construction. In the repeated graph, the invalid unsplit LTD expression produces `NaN` and therefore zero digits, while CFF and hybrid retain two tracked digits in all three numerator scenarios. The result supports using the stability mode as a representation-level diagnostic: it highlights where algebraically equivalent JSON structures differ in numerical conditioning even when the ordinary double-precision evaluator returns matching values.

Bibliography

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[2] Z. Capatti et al., causal/CFF representation construction, arXiv:2211.09653.

[3] Z. Capatti et al., further CFF representation and box examples, arXiv:2311.14374.

[4] `cLTD` and `gammaLoop` implementations, https://github.com/alphal00p/cLTD and https://github.com/alphal00p/gammaloop.